



OPEN Systematic exposure bias in city-level air quality reporting revealed by 38 years of Indian observations

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Accurately representing urban air quality remains a central challenge in managing global exposure and communicating environmental risk. In rapidly urbanising regions such as India, Air Quality Index (AQI) reporting often masks intra-city variability, limiting both scientific interpretation and public relevance. Here we analyse a 38-year air-quality record (1987–2024) from 64 consistently monitored Indian cities to quantify long-term deterioration, identify shifts in dominant pollutants, and assess the representativeness of existing AQI computation. The share of Poor–Severe days has risen from 15% to over 35%, with degradation spreading from the Indo-Gangetic Plain to southern and coastal regions. Particulate matter remains dominant in over 85% of observations, while sulphur dioxide influence has declined by 95% and nitrogen dioxide has doubled. A consultation with 450 stakeholders across 23 States revealed strong support for exposure-based reporting. We show that conventional averaging can misrepresent exposure by up to ten percentage points and introduce an exposure-weighted framework that corrects this distortion, improving the fidelity, transparency, and global transferability of city-scale air-quality assessment.

Keywords Air Quality Index (AQI), Urban Air Pollution, Exposure Assessment, Bias Correction, Environmental Governance, India

Air pollution is one of the world's most persistent environmental health challenges, responsible for an estimated 8.7 million premature deaths annually, a toll comparable to that of poor nutrition and greater than that of malaria or HIV/AIDS¹. Across regions from East Asia² and the Indo-Gangetic Plain³ to rapidly urbanising parts of Africa⁴ and Latin America⁵, chronic exposure to fine particulate matter has become a defining marker of environmental inequality⁶. While substantial research has focused on emission sources, atmospheric chemistry, and health burdens, the way air-quality information is generated and communicated remains an under-examined determinant of environmental governance^{7,8}.

India exemplifies this global tension between measurement and communication. Rapid urbanisation, industrialisation, and fossil-fuel dependence have produced some of the world's highest pollution levels^{9,10}. The country's regulatory system relies on three complementary pillars: ground-based monitoring, satellite observation, and emission-inventory modelling. Among these, ground-based stations provide the benchmark for compliance and public reporting. As of March 2025, the Central Pollution Control Board (CPCB) operates approximately 1,500 stations under two programmes: the National Air Monitoring Programme (NAMP, established in 1987) and the Continuous Ambient Air Quality Monitoring Stations (CAAQMS). Together they offer broad temporal coverage but remain spatially uneven, with dense clusters in industrial and residential zones and limited representation in commercial, sensitive, or peri-urban regions¹¹.

To improve public accessibility, the National Air Quality Index (NAQI) was introduced in 2015. It converts pollutant concentrations into sub-indices for eight regulated pollutants (PM₁₀, PM_{2.5}, NO₂, SO₂, CO, O₃, NH₃, and Pb) and reports the highest sub-index as the overall AQI¹². For city-level reporting, individual station AQIs are averaged to yield a single figure. Although this method is simple and transparent, it assumes that all monitoring sites are equally representative of population exposure. In practice, cities differ widely in the spatial mix of industrial, commercial, and residential areas, making this assumption unrealistic. As a result, city-level averages can misrepresent true exposure. A few highly polluted industrial stations can inflate the reported AQI, while an over-representation of cleaner residential sites may understate risk. In large metropolitan regions such as Delhi NCR, where more than 80 stations span over 30,000 km², a single composite value compresses substantial

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intra-city variation¹³. Thus, a communication tool designed to enhance transparency can inadvertently obscure local inequalities and reduce interpretive credibility.

International frameworks demonstrate diverse approaches to this problem. The United States Environmental Protection Agency's AQI (<https://www.airnow.gov/aqi/aqi-basics/>), Canada's Air Quality Health Index (<https://www.canada.ca/en/environment-climate-change/services/air-quality-health-index/>), and China's national AQI (www.mee.gov.cn) vary in pollutant weighting and health linkage, yet few studies have evaluated how index architecture and monitoring distribution affect representativeness^{14–16}. Technical advances in monitoring often outpace the institutional mechanisms that interpret and communicate data¹⁷, highlighting the need for frameworks that integrate scientific accuracy, fairness, and transparency.

Long-term monitoring datasets now allow such evaluation. India's NAMP and CAAQMS archives provide decades of consistent observations that can be used to assess how aggregation logic influences city-level AQI representation. Previous national assessments have examined temporal trends^{18–21} and pollutant dominance^{3,22} but have not addressed how methodological design affects representativeness or bias. Moreover, reforms to the AQI system have seldom incorporated stakeholder perspectives, even though the credibility and policy relevance of air-quality information depend on how it is interpreted by regulators, experts, and the public. Transparent and representative AQI communication is also central to Sustainable Development Goal 11.6.2 and to the objectives of India's National Clean Air Programme²³.

Despite growing reliance on AQI for policy and communication, few studies globally have empirically quantified how computational design influences representativeness. This study addresses these gaps by combining 38 years (1987–2024) of harmonised ground-based observations from 64 consistently monitored Indian cities with nationwide stakeholder consultation to evaluate and refine city-level AQI computation. Specifically, the study (i) analyses multi-decadal trends in AQI categories and dominant pollutants, (ii) surveys more than 450 experts, regulators, and citizens to identify consensus on exposure-based reporting, and (iii) tests an alternative, station-type-based framework across twelve representative multi-station cities. The comparative assessment quantifies the magnitude and direction of bias introduced by current averaging methods and evaluates the effectiveness of exposure-weighted computation in reducing misrepresentation.

By situating AQI design within a data-driven and participatory framework, this work demonstrates how modest methodological refinement can significantly improve the interpretive accuracy of air-quality reporting. The resulting exposure-weighted approach offers a transparent and transferable model for enhancing the credibility and fairness of environmental information systems in rapidly urbanising regions worldwide.

Results

Long-term variation in air-quality categories

The 38-year record (1987–2024) from 64 consistently monitored Indian cities reveals a persistent and spatially expanding deterioration in air quality (Fig. 1). Table 1 shows the details of the 64 stations and region wise distribution of States and Union Territories. The share of days falling in the Poor–Severe categories under the NAQI increased steadily from about 15% in the late 1980s to nearly 35% in recent years, while Good–Moderate days declined from 85 to 65%. This shift marks a structural transition in India's urban atmosphere from episodic exceedances to chronic, region-wide pollution.

Block I (1987–1993): limited deterioration and sparse coverage

During the early monitoring phase, air quality across most Indian cities remained within acceptable limits. Twenty-five cities, including Chandigarh, Lucknow, Patna, and Haldia, recorded no Poor or worse days in any season. Industrial centres such as Angul, Nagda, and Rourkela also exhibited sub-critical concentrations, suggesting either adequate atmospheric dispersion or the limited representativeness of the sparse network.

Early signs of degradation appeared mainly during the winter and post-monsoon seasons, driven by stagnation and low mixing heights. Nine cities reported more than 10% of winter days in the Poor or worse categories, predominantly across the Indo-Gangetic Plain (IGP). The monsoon season remained largely clean, with Poor days restricted to isolated cases such as Dehradun (54%), Delhi (3%), and Howrah (1%). Overall, this period reflected the initial stages of urban-industrial expansion before large-scale emissions growth.

Block II (1994–2008): network stabilisation and emerging deterioration

By the mid-1990s, improved network coverage began capturing spatial variability more effectively. Air quality declined across most regions, coinciding with accelerated urbanisation and industrialisation. Cities such as Delhi (33% of winter days), Agra (18%), and Ghaziabad and Howrah (17%) recorded a growing frequency of Poor–Severe days. Industrial hubs like Angul, Ankleshwar, and Vapi remained relatively stable but showed increasing intra-annual variability.

The rise in Poor air-quality days during this period marks the onset of structural deterioration rather than isolated exceedances. Summer dust episodes in arid cities such as Jaipur and Jodhpur, coupled with stagnant winter conditions in the IGP, contributed to this trend. The monsoon season continued to provide transient relief but no longer offset the overall annual deterioration.

Block III (2009–2024): dense monitoring and widespread exceedances

The introduction of the CAAQMS network after 2017 dramatically increased data density and spatial coverage. This expansion exposed the full extent of degradation that had previously remained under-detected. No city recorded zero Poor days in this period. During post-monsoon months, 23 cities reported more than 30% of days in the Poor–Severe range.

The Delhi–NCR region remained the most polluted cluster, with Delhi (61%), Noida (58%), Ghaziabad (54%), and Faridabad (52%) showing the highest frequency of degraded days. Importantly, deterioration was no

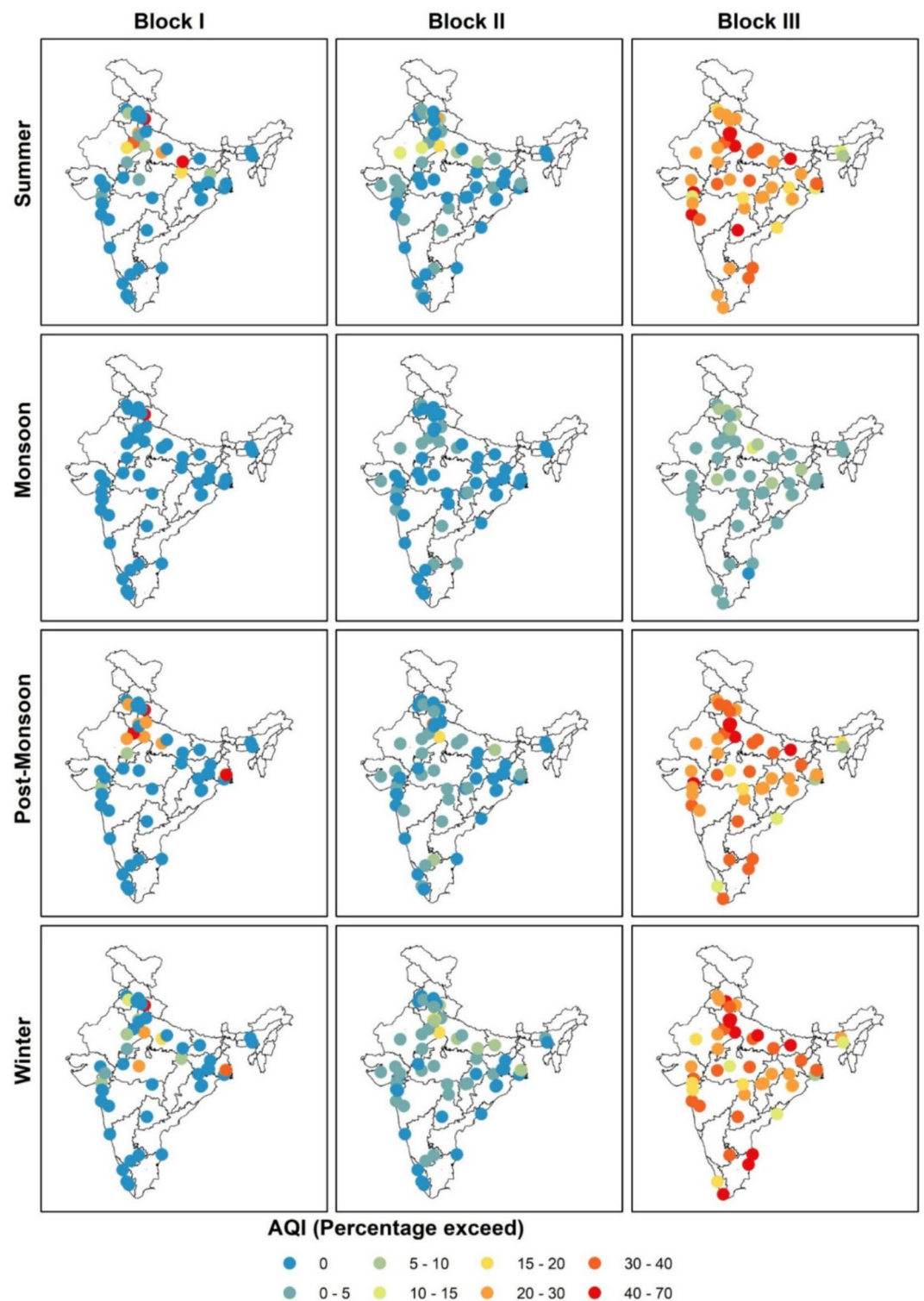


Fig. 1. Expansion of Poor–Severe Air Quality Days across 64 Indian Cities by Season and Monitoring Period (Block I: 1987–1993; Block II: 1994–2008; Block III: 2009–2024).

Region	State/Union Territory	City
Indo-Gangetic Plains	Bihar	Patna
	Chandigarh	Chandigarh
	Delhi	Delhi
	Haryana	Faridabad, Yamunanagar
	Jharkhand	Dhanbad, Jamshepur, Sindri
	Punjab	Ahmedgarh, Ludhiana, Jalandhar
	Uttar Pradesh	Agra, Anpara, Gajraula, Ghaziabad, Kanpur, Lucknow, Noida, Varanasi
	Uttarakhand	Dehradun
	West Bengal	Haldia, Howrah, Kolkata
Central	Chhattisgarh	Bhilai, Korba, Raipur
	Gujarat	Ahmedabad
	Maharashtra	Chandrapur, Mumbai, Nagpur, Pune
	Madhya Pradesh	Bhopal, Indore, Nagda, Jabalpur
	Odisha	Angul, Rourkela, Talcher
North West	DNH & DD	Daman, Dadra Nagar Haveli, Silvassa
	Gujarat	Ankleshwar, Rajkot, Surat, Vadodara, Vapi
	Rajasthan	Alwar, Jaipur, Jodhpur, Kota
South	Andhra Pradesh	Visakhapatnam
	Karnataka	Bengaluru, Mysuru
	Goa	Panaji
	Kerala	Kochi, Kottayam, Kozhikode, Thiruvananthapuram
	Puducherry	Puducherry
	Tamil Nadu	Chennai
	Telangana	Hyderabad
North East	Assam	Guwahati
	Meghalaya	Shillong
Himalaya	Himachal Pradesh	Shimla, Paonta Sahib, Parwanoo

Table 1. Consistently monitored indian cities and regional distribution (1987–2024).

longer confined to the IGP. Southern and coastal cities such as Chennai (35%), Hyderabad (33%), and Mumbai (37%) now displayed frequent exceedances, indicating the diffusion of pollution stress to regions previously considered clean. Central Indian cities including Indore (34%) and Jabalpur (31%) showed similar trajectories.

National and seasonal patterns

The long-term dataset highlights two consistent features: (i) winter remains the dominant pollution season due to low boundary-layer heights and biomass-burning influences, and (ii) regional gradients persist despite nationwide deterioration. The IGP, Central, and Northwestern regions experience the highest proportion of Poor–Severe days, while the South and Northeast remain comparatively cleaner, supported by favourable meteorology and lower emission densities.

Across 38 years, India's urban air-quality trajectory has transformed from spatially localised degradation to nationwide persistence. The expansion of monitoring coverage has improved detection but also confirmed genuine environmental decline linked to increased energy use, industrial activity, vehicular emissions, and construction dust. Together, these findings depict a long-term structural shift where chronic exposure, rather than short-term peaks, defines urban air quality.

Transitions in dominant pollutants

Figure 2 presents the dominant pollutant during the winter season across the three observational blocks.

Block I (1987–1993): localised particulate dominance

During the initial monitoring period, PM_{10} emerged as the dominant pollutant in 48 cities during winter, while SO_2 dominated in Jamshedpur and NO_2 in a few commercial centres. Twenty cities recorded PM_{10} as the dominant pollutant on all winter days, covering most regions except the Himalaya and Northeast. The highest frequency of PM_{10} dominance was observed in Vadodara (105 days), Howrah (80 days), Delhi (80 days), Surat (68 days), and Nagpur (56 days). These cities represented a mix of industrial and traffic-impacted environments with limited dispersion capacity.

NO_2 was the second most frequent dominant pollutant, particularly in Kota, Guwahati, and Jalandhar, where monitors were located in densely commercial zones with high vehicular activity. SO_2 dominance was confined to a small cluster of 13 industrial cities, most notably Jamshedpur, reflecting the influence of metallurgical and thermal power sources. Across other seasons, PM_{10} remained dominant except at isolated stations in Talcher (summer) and Jalandhar and Shillong (post-monsoon and monsoon).

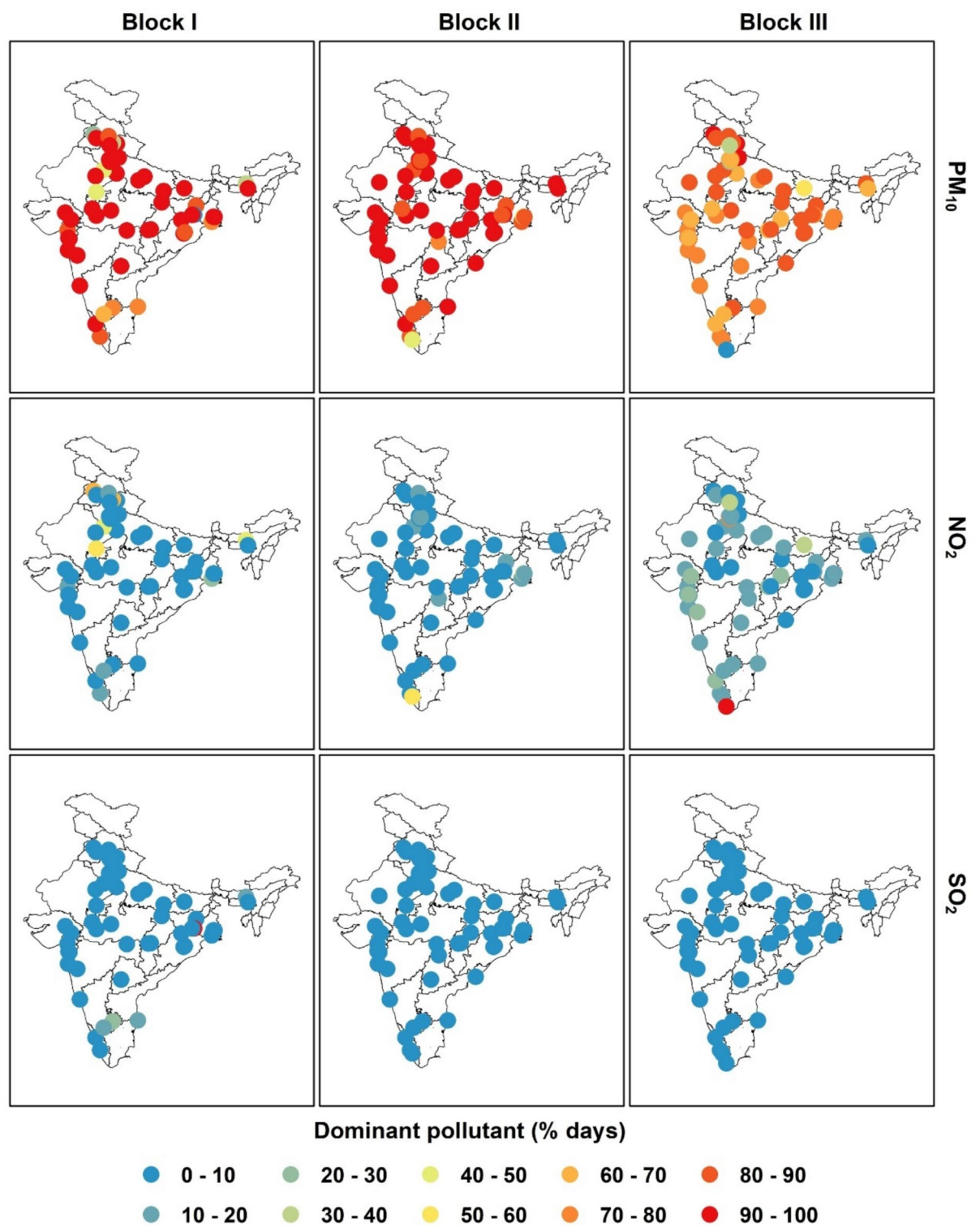


Fig. 2. Dominant air pollutants (PM₁₀, NO₂, SO₂) across Indian cities over three monitoring phases (Block I 1987–1993, Block II 1994–2008, and Block III 2009–2024).

This period represents the early stage of particulate accumulation in Indian cities. The limited spatial coverage of the network masked regional heterogeneity but revealed the first indications of PM₁₀ prevalence in industrial corridors and urban transport hotspots.

Block II (1994–2008): expansion and consolidation of particulate dominance

During this period, PM₁₀ became the overwhelmingly dominant pollutant nationwide, identified in nearly all cities during winter. Eleven cities, including Agra, Ghaziabad, Gajraula, and Varanasi in the IGP, and Goa and Kozhikode in southern India, reported PM₁₀ as the dominant pollutant on every monitored day. Very high frequencies of PM₁₀ dominance were recorded in Delhi (98% of days), Hyderabad (96%), Guwahati (97%), Ahmedabad (98%), Jaipur (93%), and Kanpur (99%).

NO₂ emerged as the secondary dominant pollutant, appearing in 49 cities, particularly in Howrah, Kottayam, and Chandrapur, where transport and industrial combustion were prominent. SO₂ remained dominant in 21 cities, mainly in metallurgical and power-generation centres such as Chandrapur, Haldia, Nagda, Sindri, and Jamshedpur, though its overall frequency was low compared with PM₁₀.

Seasonal patterns showed that PM₁₀ dominance persisted year-round, including during the monsoon, while NO₂ occasionally emerged during wetter months due to reduced dust concentrations from wet deposition. The relative increase in SO₂ dominance during the monsoon likely reflects lower particulate interference in index computation.

This decade marked the transition from localised to widespread particulate dominance, coinciding with rapid urban growth and industrial intensification. The persistence of PM₁₀ across all regions indicates that emission growth was outpacing regulatory control, highlighting the limitations of single-pollutant management strategies.

Block III (2009–2024): network expansion and emerging multi-pollutant regime

The third block corresponds to a transformative phase in India's monitoring network, driven by the large-scale deployment of CAAQMS and the inclusion of new parameters under the National Ambient Air Quality Standards (2009). Within the 64 consistently monitored cities, PM₁₀, SO₂, and NO₂ remained the common parameters for long-term comparison.

Across all cities, PM₁₀ continued to dominate, accounting for 86.1% of total AQI-dominant days, followed by NO₂ (12.8%) and SO₂ (0.9%). This enduring particulate dominance reflects the combined effects of dust resuspension, construction activity, and combustion emissions that continue to shape India's urban air quality.

Cities such as Kolkata, Lucknow, Jaipur, and Nagpur recorded more than 80% of winter days with PM₁₀ as the dominant pollutant. Delhi exhibited a modest decline in PM₁₀ dominance (approximately 81%), accompanied by a relative rise in NO₂, indicating growing contributions from vehicular emissions and secondary nitrogen chemistry.

SO₂ dominance declined markedly, appearing on fewer than 1% of AQI dominant days, even in major industrial centres such as Chandrapur, Haldia, and Ludhiana. This reduced influence of SO₂ within the AQI framework may, in part, reflect the increasing dominance of particulate matter, whereby PM-based sub-indices overshadow SO₂ contributions in index computation. Long-term concentration trends indicate a concurrent reduction in ambient SO₂ levels, as documented in our companion analysis²⁴. However, the present results do not allow definitive attribution of declining SO₂ dominance to specific regulatory measures alone, and any role of emission controls or fuel transitions is therefore interpreted cautiously as a potential contributing factor.

Together, these patterns signify a transition from a coarse particulate–dust regime toward a more complex multi-pollutant atmosphere. Technological expansion in monitoring has revealed these emerging patterns, while the continued dominance of particulate matter underscores persistent structural emissions linked to construction, transport, and energy use.

Stakeholder and expert perspectives on city-level AQI reporting.

To complement the empirical analysis of long-term air-quality data, a nationwide stakeholder and expert survey was conducted to evaluate how air-quality information should be computed and communicated in multi-station cities. The survey linked technical aspects of AQI design with dimensions of transparency, representativeness, and interpretability. Between October 2024 and May 2025, 450 valid responses were collected from 23 States and Union Territories, comprising 94 (21%) participants from academia and research institutions (Category A), 284 (63%) from regulatory agencies, industries, and NGOs (Category B), and 72 (16%) from the general public (Category C). Among respondents, 74% were male and 26% female; 71% were below 45 years, 24% between 46–60 years, and 5% above 60 years.

Preferences regarding monitoring locations

Stakeholders were first asked to identify which type of monitoring site should be prioritised when communicating city-level AQI. Across all respondents, residential locations were considered most relevant by 63%, followed by industrial (27%) and sensitive zones (9%).

Category-wise analysis revealed consistent patterns. Among academicians and researchers (Category A), 72% prioritised residential stations, 22% industrial, and 5% sensitive sites. Health experts within this group assigned relatively higher importance to industrial zones (36%), reflecting occupational exposure concerns. In Category B, which included regulators and industry representatives, 62% favoured residential locations, 16% industrial, and 7% sensitive zones. Within this group, officials from Pollution Control Boards preferred residential sites (66%), while industry respondents showed a more balanced distribution between residential (56%) and industrial (34%) sites. Among the general public (Category C), 57% prioritised residential areas, 35% industrial, and 8% sensitive zones.

The convergence across respondent groups underscores a shared perception that AQI reporting should reflect population exposure in living environments rather than proximity to emission sources. This finding

highlights an emerging consensus among both regulators and citizens on aligning air-quality communication with exposure relevance.

Preferred methodology for city-level AQI reporting

Respondents were also asked to identify the most appropriate computation approach for city-level AQI in multi-station cities (Fig. 3). A majority (52%) supported disaggregated reporting, recommending that separate AQI values be published for industrial (AQI_{ind}), residential (AQI_{res}), and sensitive (AQI_{sen}) areas. An additional 24% preferred that AQIs be computed separately for each station type but also combined into an overall average. Twelve percent supported a “maximum-of-maximum” approach, where the highest AQI across all stations defines the city-level value. Only 7% endorsed the current CPCB method of a single averaged AQI, and 4% favoured a minimum-based approach. These options were intentionally limited to approaches that are administratively feasible and implementable using the existing monitoring infrastructure of Pollution Control Boards. Geographically resolved, sub-regional AQI reporting was not evaluated, as most Indian cities do not yet have sufficiently dense or spatially uniform monitoring networks to support such reporting on an operational basis.

Regulatory respondents showed the strongest support for reform. Among Pollution Control Board officials, 54% preferred disaggregated reporting, and only 5% supported the existing averaging approach. A similar trend was observed among academicians (55%), while health experts showed even stronger support (73%) for separate AQI reporting by station type.

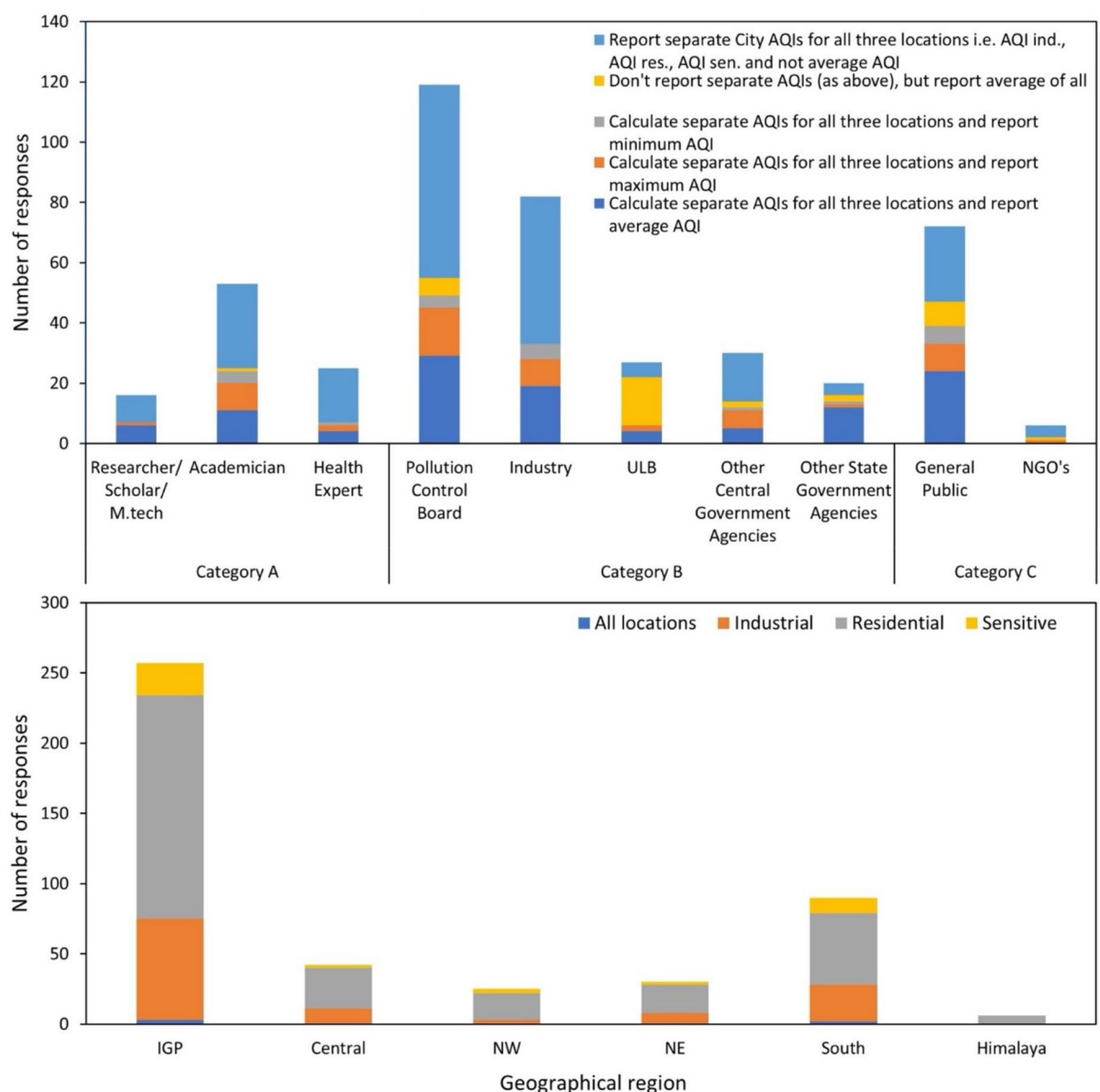


Fig. 3. Preferred air quality index (AQI) reporting method across professional groups and regions.

Overall, respondents viewed the single-number average as a source of information loss that conceals intra-city gradients and may misrepresent exposure risks. These perspectives are consistent with practices adopted in China's zonal AQI framework and Canada's Air Quality Health Index, both of which employ spatial or health-risk weighting. The broad support for disaggregated reporting in India aligns with global trends toward exposure-representative metrics and supports the transparency objectives outlined under Sustainable Development Goal (SDG) indicator 11.6.2.

Regional and demographic variation

Responses were geographically distributed across six regions: Indo-Gangetic Plain (n=257), South (n=90), Central (n=42), Northeast (n=30), Northwest (n=25), and Himalaya (n=6) (Fig. 5). Across all regions, residential monitoring sites were most frequently prioritised, accounting for 61.9% of respondents in the IGP, 56.7% in the South, 69.0% in Central India, 66.7% in the Northeast, 76.0% in the Northwest, and 100% in the Himalaya. Industrial zones ranked second in all regions, averaging 26–29% preference, while sensitive locations consistently received below 10%.

Gender-based analysis yielded similar trends. Among female respondents, 73 selected residential, 29 industrial, and 14 sensitive locations. Among males, 209 chose residential, 91 industrial, and 27 sensitive zones. This uniformity across regions and demographics indicates a strong and consistent national orientation toward residential representativeness as the reference point for environmental communication.

Comparable convergence has been observed in participatory air-quality initiatives in Latin America and East Asia, where citizen input increasingly influences network design. The results from India reflect a similar progression, suggesting that public and institutional expectations are converging toward transparency and social relevance in air-quality reporting.

Evaluation of city-level AQI reporting frameworks

The percentage of days in the Poor–Severe AQI categories derived using three computation approaches: (i) the CPCB's official city-average method, (ii) a residential- only scheme, and (iii) an industrial and other-station grouping was analysed. The analysis covers twelve representative cities, i.e. five major (Delhi, Bhopal, Chennai, Guwahati, and Ahmedabad) and seven industrial (Singrauli, Angul, Talcher, Visakhapatnam, Ankleshwar, Baddi, and Byrnihat), across three temporal blocks. Table 2 shows the classification of different stations in those cities. In Delhi, Chennai, and Ahmedabad, the number of monitoring stations in industrial and residential locations is roughly comparable, with a ratio of approximately 1:1.2. In contrast, Guwahati and Bhopal have higher number of stations located in residential and commercial areas, and very few stations in industrial locations. Historically, major cities have maintained air quality monitoring stations across all location categories, including residential, industrial, and commercial (or other) areas. All selected industrial cities have multiple air quality monitoring stations in the residential and industrial locations. Angul, Ankleshwar, Baddi, and Talcher each have two stations located in residential areas and two in industrial areas. Monitoring in Brynihat comprises two stations in industrial zones and one in a residential area. However, Singrauli has three residential stations and one industrial station situated at Northern Coalfield Limited. Visakhapatnam a major industrial and commercial hub in the south region has 11 monitoring stations, six in residential areas, three in industrial site, one in a commercial area, and one in sensitive location. Figure 4 shows the results for Block III.

Block I: sparse networks, negligible divergence

Monitoring networks in this period were rudimentary, with one or two centrally located stations per city. Across all eight cities with data, the median absolute difference between averaged and residential Poor–Severe shares was nearly zero, and the same held for the industrial + others comparison. Typical differences were < 1 percentage point, confirming that the single averaged AQI adequately reflected city-wide conditions when monitoring was limited and spatial variability was small.

Cities	Manual Stations			Continuous Ambient Air Quality Monitoring Stations		
	Residential	Industrial	Other	Residential	Industrial	Other
Delhi	7	4	1	18	14	7
Ahmedabad	7	4	0	4	2	3
Bhopal	7	1	2	1	0	2
Angul	2	1	0	0	0	1
Talcher	1	2	0	0	1	0
Ankaleshwar	1	1	0	0	1	0
Visakhapatnam	5	3	1	0	0	1
Chennai	7	6	0	3	4	2
Guwahati	7	0	0	2	0	2
Singrauli	2	1	0	1	0	0
Baddi	1	1	0	0	0	1
Byrnihat	1	0	0	0	0	1

Table 2. Monitoring station distribution across twelve major and industrial Indian cities.

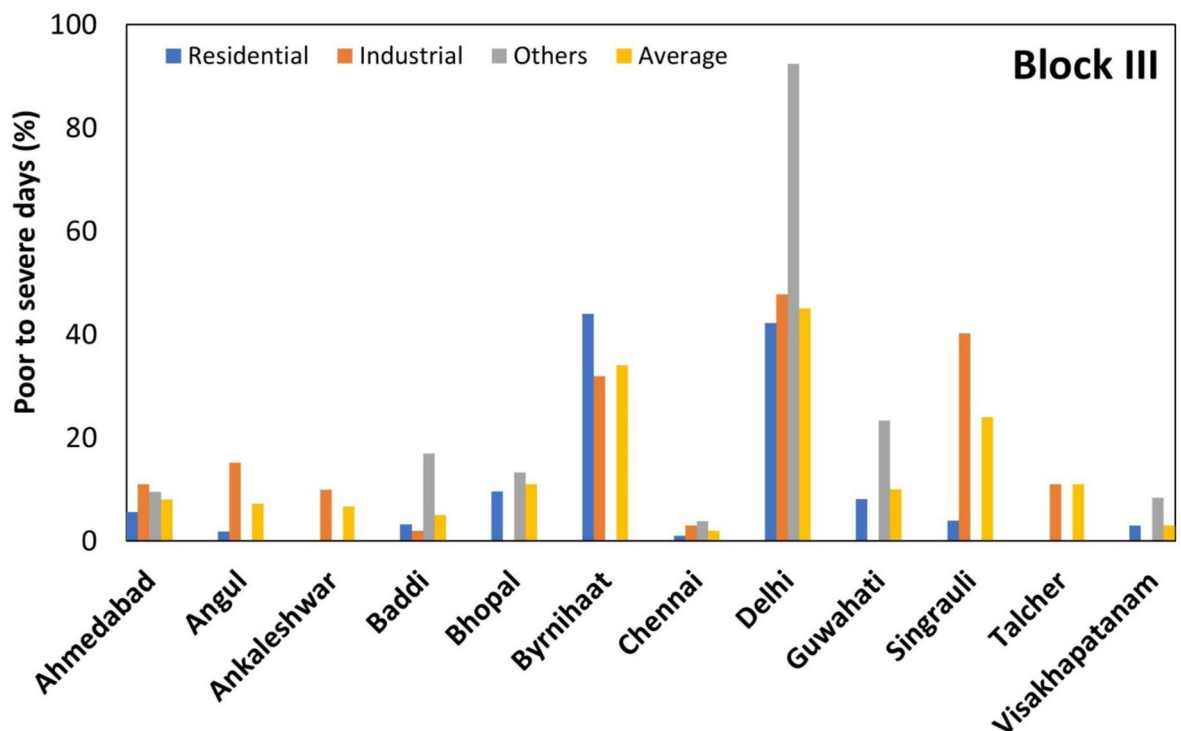


Fig. 4. Comparison of poor–severe air quality days across monitoring location types and city-Average AQI in twelve representative Indian cities (Block III: 2009–2024).

Block II: expanding coverage and emerging bias

With the growth of NAMP stations and the onset of network diversification, systematic discrepancies began to appear. The median absolute difference between averaged and residential Poor–Severe shares rose to ~0.02 percentage points, while that between averaged and industrial + others reached ~0.08. Mean deviations, though modest (~1.6 points), reveal the early onset of representativeness bias.

Industrial towns displayed the most evident divergence. In Baddi, the averaged AQI indicated 15.7% of days in Poor–Severe categories compared with only 3.4% for residential locations, a ~12-point overestimation of population exposure. Among major cities, Delhi (16% vs 9.9%) recorded maximum differences. Although small, these emerging gaps foreshadowed stronger distortions once continuous monitoring expanded.

Block III: dense networks, amplified divergence

During the most recent decade, parallel operation of NAMP and CAAQMS stations substantially increased spatial heterogeneity. The median absolute difference between city-average and residential Poor–Severe shares reached 2.6 percentage points, and between city-average and industrial and others 2.2 points.

In all major cities, residential AQI is less than the average. For example, in Delhi, 45% of days were Poor–Severe by the averaged method, compared with 42% (residential). In Ahmedabad it was 8% (average) compared to 5.5% (residential). Similar conclusions were achieved from all industrial cities except Byrnihaat. For example, in Singrauli a severely polluted area in the IGP known for coal mining and thermal power plants, it was 4% (residential) compared to 24% (average). Data from the industrial site (Northern Coalfields Ltd.) indicated significantly higher pollution levels compared to residential areas. Byrnihaat displayed the widest underestimation, where residential monitors recorded 43.9% Poor–Severe days versus 34% from averaging.

Overall, across all cities, the direction and magnitude of deviation were shaped by the spatial mix of monitoring stations. Where industrial monitors dominate, the averaged AQI inflates city-wide pollution levels; where residential monitors are fewer or located in cleaner zones, the average under-represents population exposure. From Block I to III, the typical absolute deviation between averaged and location-specific AQIs increased from <1% to >10%, confirming that enhanced data density alone does not ensure representativeness. These results empirically substantiate stakeholder feedback favouring disaggregated AQI publication and demonstrate that station-type-specific reporting or exposure-weighted aggregation is necessary to preserve the scientific and governance integrity of India's air-quality information system.

The results of the comparative assessment converge toward a revised city-level AQI reporting framework for multi-station cities. Specifically, we recommend reporting AQI separately by monitoring station type (AQI_{ind}, AQI_{res}, AQI_{sen}), rather than as a single city-wide averaged AQI. This station type based approach explicitly accounts for land-use heterogeneity within cities and provides a more representative and interpretable assessment of city-level air quality than conventional averaging across all monitoring locations.

Discussion

Accurate representation of urban air quality is essential for both scientific interpretation and environmental governance, yet the representativeness of city-level reporting remains weakly examined. Conventional AQI computation aggregates data from heterogeneous stations without accounting for population exposure or land-use context, potentially distorting risk communication and policy response. This study addresses that gap by analysing a 38-year air-quality record from 64 Indian cities to identify pollutant transitions, quantify bias in current AQI methods, and propose an exposure-weighted reporting framework. The findings illuminate systemic changes in India's atmospheric composition and reveal computational distortions with broad implications for global monitoring practice.

India's long-term record shows a steady and spatially expanding deterioration in urban air quality. The proportion of Poor–Severe AQI days has more than doubled since the late 1980s, with degradation extending from the Indo-Gangetic Plain to southern and coastal regions. Cities once considered clean now experience chronic exceedances, indicating that poor air quality has become a structural feature of India's urban atmosphere. Persistent winter stagnation and post-monsoon peaks suggest that meteorology amplifies continuous emissions rather than producing episodic events. These results confirm that India's pollution problem is entrenched and multifactorial.

Particulate matter remains the dominant pollutant across most cities, reflecting emissions from construction, industry, and transport coupled with dust resuspension and secondary formation. The doubling of nitrogen dioxide dominance indicates a transition toward combustion-linked sources, while the disappearance of sulphur dioxide dominance reflects cleaner fuels and process controls. Together, these changes signal India's shift from a coarse, dust-driven regime to a finer, chemically complex atmospheric system, a trajectory comparable to other industrialising economies but one that demands multi-pollutant management rather than single-species control.

The nationwide consultation across 23 States offers additional perspective on how air-quality information is valued. Among 450 respondents, including regulators and citizens, there was broad consensus that AQI reporting should represent exposure rather than emission intensity. Residential monitoring sites were prioritised by 63% of respondents, and 52% supported disaggregated publication by station type. This alignment between expert and public preferences underscores growing recognition that transparency and representativeness are integral to credible environmental data.

Comparative evaluation of twelve multi-station cities quantifies how computation design influences AQI outcomes. The current CPCB averaging method assumes uniform representativeness, producing systematic bias as networks expand. When industrial monitors dominate, city averages overstate exposure; when residential stations are sparse or cleaner, exposure is understated. In Singrauli and Talcher, averaged AQI values exceeded residential estimates by up to ten percentage points, whereas in Bijnor the reverse occurred. Such discrepancies show that expanding monitoring coverage without revising aggregation logic does not improve representativeness. Aggregation design thus emerges as a methodological determinant of both scientific accuracy and policy interpretation.

The exposure-weighted, station-type-based framework proposed here addresses this distortion by incorporating land-use representativeness into computation. It reduces bias, stabilises inter-city rankings, and can be implemented without additional infrastructure. Integrating such computation into routine reporting would enhance interpretive precision and align with SDG 11.6.2, which calls for transparent and equitable air-quality data. The approach is scalable and transferable to other data-rich but spatially heterogeneous monitoring networks worldwide.

These insights also have implications for region-wide response plans such as the Graded Response Action Plan (GRAP) in the Delhi–NCR region. Although GRAP provides a structured mechanism for intervention, its effectiveness depends on the representativeness of the AQI used to trigger action. At present, GRAP relies on a single city-level AQI derived from averaging AQI values across all monitoring stations, which can obscure local contrasts across heterogeneous land-use zones and result in uniform restrictions that may not be appropriate for all areas. While station-level AQIs are already publicly disseminated, implementing GRAP actions independently for each monitoring location, particularly in large metropolitan regions with dense monitoring networks, poses significant operational and administrative challenges. Within this practical context, reporting separate AQIs for industrial, residential, and ecologically sensitive zones, based on the existing CPCB classification of monitoring stations, offers a feasible intermediate refinement. This approach improves representativeness relative to a single city-average AQI while retaining operational simplicity and enabling more targeted interventions aligned with dominant pollution sources across different land-use zones.

The findings also extend current literature. Previous national assessments have documented pollution trends but rarely examined how computational design shapes exposure interpretation. By combining long-term empirical analysis with stakeholder insight, this study demonstrates that methodological refinement can enhance both scientific reliability and public trust. Similar approaches are emerging in the European Union²⁵, the United States²⁶, East Asia, Africa and Latin America²⁷, where exposure-oriented frameworks increasingly underpin air-quality governance.

This analysis is constrained by the availability of consistent long-term pollutant data. Parameters such as PM_{2.5} and ozone were excluded from historical comparison due to limited temporal continuity, and stakeholder responses may overrepresent technically informed participants. Further, the present study does not account for population density, activity patterns, or the actual time spent by individuals across different land-use areas. The land-use categories adopted, industrial, residential, and sensitive, are based on the CPCB's classification of manual and continuous monitoring stations and the land-use definitions specified in the NAAQS, 2009. Consequently, the analysis reflects relative exposure potential across land-use types rather than actual population exposure.

In conclusion, India's 38-year record reveals a shift from episodic to chronic urban pollution dominated by particulates and nitrogen compounds. Aggregation bias has become a significant interpretive limitation in AQI

reporting. The proposed exposure-weighted computation offers a scientifically robust and operationally feasible correction that improves representativeness, fairness, and public confidence. Accurate measurement paired with equitable computation is fundamental to credible air-quality communication and more effective atmospheric management.

Methods

Integrated methodological framework (Fig. 5) underpinning this study comprises three interconnected components: (i) empirical analysis of historical monitoring data; (ii) structured stakeholder and expert consultation, and (iii) comparative evaluation of revised AQI computation schemes. Together, these components link technical recalibration with governance-oriented interpretation, aligning scientific credibility with transparency and representativeness in policy communication.



Fig. 5. Integrated methodological framework combining long-term empirical analysis with stakeholder-informed AQI reform.

Data sources and historical AQI analysis

This study builds on the national air-quality dataset used in our earlier analysis of long-term air-quality trends across Indian cities during 1987–2019²⁴. That work comprehensively detailed the data architecture, instrumentation, and quality-assurance protocols associated with the NAMP and the CAAQMS network operated by the CPCB. The present study extends this rigorously quality-controlled database both temporally and analytically to examine how India's evolving monitoring network can inform reform in city-level AQI methodology.

Together, NAMP and CAAQMS constitute one of the world's largest regulatory air-quality monitoring systems, encompassing 974 manual stations and 558 continuous stations across 420 cities in 28 States and 7 Union Territories. In total, the dataset contains more than 10.1 lakh daily observations from NAMP (1987–2024) and 7 lakh daily records from CAAQMS (2007–2024). While spatial coverage has expanded rapidly since 2017, earlier years show considerable variability in data completeness.

For inter-decadal comparability, the analysis focuses on 64 cities that maintained stable monitoring locations and data continuity for at least three decades. These “mostly consistently monitored cities” were identified using three criteria: (i) $\geq 70\%$ annual data completeness, (ii) continuity across all three pollutants, and (iii) availability of validated observations from both manual and continuous stations. Cities were grouped into six regional classifications, i.e. Himalaya, Indo-Gangetic Plain, Northwest, Central, South, and Northeast, following Guttikunda, et al.²⁸.

Rigorous data-cleaning procedures were applied in accordance with CPCB's Guidelines²⁹. Anomalous entries, such as consecutive identical readings, zeros, negatives, or concentrations beyond detection limits, were excluded. For analytical comparability, data transformations, pollutant conversions, and the regional classification of cities into six zones follow the methodology described in Sharma, et al.²⁴. The analysis spans three multi-year blocks that reflect major policy and network phases: (i) 1987–1993: Early NAMP expansion; (ii) 1994–2008: Network consolidation and urban intensification; (iii) 2009–2024: CAAQMS integration and post-NAAQS 2009 reforms including National Clean Air Programme.

The NAQI was introduced in 2015 primarily as a public communication instrument and is structurally anchored to the NAAQS, 2009. While India's ambient air quality standards were revised in 1982, 1994, and 2009, the present analysis consistently applies NAQI breakpoints aligned with NAAQS, 2009 for all years considered. This approach ensures comparability across monitoring blocks and prevents inconsistencies associated with evolving standards and the addition or removal of regulated parameters over time. For each block, city-level annual averages were used to compute NAQI. Sub-indices were calculated for individual pollutants, and the highest sub-index defined the overall AQI category, Good, Satisfactory, Moderately Polluted, Poor, Very Poor, or Severe. The pollutant producing the highest sub-index was designated the dominant pollutant, allowing analysis of temporal shifts in India's principal air-quality drivers. Seasonal means were also derived for summer (Apr–Jun), monsoon (Jul–Sep), post-monsoon (Oct–Nov), and winter (Dec–Mar).

By leveraging India's extensive monitoring infrastructure while isolating its most temporally consistent subset, this study combines data abundance with methodological rigour. The resulting 38-year harmonised dataset (1987–2024) provides a robust empirical foundation for assessing long-term variation in AQI categories and pollutant dominance, and for testing governance-aligned frameworks for city-level AQI computation and communication. It also supports testing of governance-aligned frameworks for city-level AQI computation and communication that are relevant to global efforts in standardising air-quality information systems.

Stakeholder and expert consultation

To evaluate the interpretability and governance relevance of the revised AQI framework and to inform the development of a more representative city-level reporting methodology, a structured stakeholder and expert consultation titled “Survey on Air Quality Index (AQI): Expert Opinion and Public Perception” was conducted between October 15, 2024, and May 31, 2025 (Table 3). The survey was designed to capture diverse perspectives on how AQI data should be computed and communicated in multi-station cities.

Participants were asked to indicate the relative importance they assign to AQI values from stations in industrial, residential, and sensitive zones, and to identify their preferred method for city-level AQI reporting, whether as a single aggregated value or disaggregated by station type. The sensitive areas considered in the present study are ‘Ecologically Sensitive Areas’ which are notified by MoEFCC. Additional items examined perceptions of clarity, fairness, and credibility in current reporting, and the perceived policy usefulness of revised approaches.

The questionnaire comprised items organised into three thematic sections:

- i. Understanding of the current AQI system
- ii. Evaluation of the revised methodology: understanding and acceptance of the station-type-based approach and practical feasibility of its adoption.
- iii. Policy relevance and reporting

A five-point Likert scale (1 = not important, 5 = very important) supplemented by open-ended prompts for qualitative feedback. The form was reviewed by five domain experts to ensure clarity and relevance, and a pilot test with 27 participants was used to refine question flow and phrasing prior to full dissemination.

The survey was circulated through Google Forms and manual data collection ensuring representation from all major regions of India. Respondents were grouped into three categories to reflect diverse expertise and engagement:

- i. Category A: Academicians, researchers, postgraduate students, and health experts;

Question	Response Options
Name	—
Age *	Under 18 / 18–29 / 30–44 / 45–59 / 60 +
State *	—
District *	—
City/Town *	—
Gender *	Female / Male / Other
Education *	Primary / Secondary / Graduate / Postgraduate / Doctorate / Other
Affiliation / Occupation *	Academia / Health and medical Expert / Research and Development/ Industry Professional/ Pollution Control Board / Urban Local Body / Non-Governmental Organization / Government Agency / Private Sector / Self-Employed / Homemaker / Other
Have you heard of the AQI? *	Yes / No
Importance of good air quality (1–5) *	1 / 2 / 3 / 4 / 5
Do you know about the national air-quality reporting system? *	Yes / No
Most important location for reporting ambient air-quality data *	Industrial / Residential / Sensitive
Preferred method for City-level AQI computation *	Separate AQIs by location / Overall average / Minimum AQI / Maximum AQI
Email address	—
Mobile number	—

Table 3. Survey questionnaire on air quality index (AQI): expert opinion and public perception. “*” indicates a required question.

- ii. Category B: Officials from the Central Pollution Control Board, State Pollution Control Boards, Pollution Control Committees, industries, government departments (including urban local bodies), and NGOs;
- iii. Category C: General public and concerned citizens.

Quantitative data were summarised using descriptive statistics (means, frequencies, and percentages), while qualitative remarks were thematically coded under governance domains of transparency, credibility, and equity. Insights from this exercise were subsequently used to design the comparative methodological evaluation, where AQI values for twelve cities across six geographical regions were calculated using both the traditional city-wide average and alternative reporting schemes derived from stakeholder inputs across the three temporal blocks. This linkage enabled the examination of how different reporting approaches could affect category assignment and ranking outcomes.

Overall, this survey constitutes one of the first nationwide efforts in India to integrate stakeholder and expert feedback into the methodological reform of an environmental information system. Earlier work on AQI has largely focused on technical refinement or public-risk communication, whereas this survey directly examines the institutional usability and governance legitimacy of methodological alternatives. Embedding stakeholder input grounds the proposed framework in regulatory reality and social acceptance, reinforcing the AQI's function as both a scientific indicator and a governance instrument for transparent and equitable air-quality communication. The participatory approach adopted here provides a transferable template for other developing and transitional economies seeking to integrate inclusivity, transparency, and evidence-based design into their air-quality management systems.

Revised AQI computation and validation

Findings from the expert–stakeholder consultation informed the design of an alternative AQI computation framework that evaluates how different city-level aggregation strategies affect reported outcomes. Two approaches were compared:

- i. Baseline method (NAQI by CPCB): the existing regulatory approach in which a single city-level AQI is computed as the mean of all station-level AQIs.
- ii. Survey-informed alternative method: a modified framework that evaluates AQI computation schemes for city based on the recommendations of the survey.

The comparative analysis was conducted for 12 representative multi-station cities across six geographical regions, reflecting India's climatic, demographic, and industrial diversity. The selection includes both major and industrial cities, chosen based on data availability from manual and real-time monitoring stations. The major cities (Delhi, Chennai, Bhopal, Ahmedabad, and Guwahati) are significant urban centres and key commercial hubs within their respective regions; all are part of the Smart City Mission of Government of India. The selected industrial cities (Singrauli, Angul, Talcher, Visakhapatnam, Ankleshwar, Baddi, and Byrnihat) host industrial clusters and/or potential impact zones identified by CPCB under the Comprehensive Environmental Pollution Index (CEPI). For each city, AQI categories and rank orders were computed under both methods across three temporal blocks (1987–1993, 1994–2008, and 2009–2024).

Sub-indices were derived from validated annual mean concentrations of pollutants using CPCB's official break-point tables. This comparative framework quantifies how aggregation logic can influence AQI categories,

dominant-pollutant identification, and city rankings. Embedding this technical recalculation within a participatory governance process reframes the AQI as a policy-relevant information architecture that connects empirical credibility with communication transparency. The framework offers transferable insights for other rapidly urbanising regions that aim to design equitable and trusted air-quality reporting systems aligned with global sustainability and air-quality management goals.

Data availability

The datasets generated and/or analysed during the current study are not publicly available due to data-sharing restrictions, but are available from the corresponding author upon reasonable request. Derived analytical outputs and computation codes used to generate figures and statistical analyses are available upon request for academic, non-commercial use.

Received: 2 November 2025; Accepted: 10 February 2026

Published online: 03 April 2026

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Acknowledgements

The authors would like to thank the Central Pollution Control Board (CPCB), Government of India, for providing access to monitoring data for this research.

Author contributions

Garima Sharma: Methodology, Data curation, Visualization, Writing- Original draft preparation. Pareshbhai Dineshbhai Parmar: Visualization, Writing- Original draft preparation. Arvind Kumar Nema: Reviewing and Editing. Prashant Gargava: Reviewing and Editing. Sri Harsha Kota: Conceptualization, Methodology, Reviewing and Editing.

Declarations

Competing interests

The authors declare no competing interests.

Ethical approval

The survey conducted in this study collected opinions from adult professionals and experts through a voluntary and anonymous online questionnaire. No personal, medical, or sensitive information was collected. All methods were carried out in accordance with relevant guidelines and regulations. Informed consent was obtained from all respondents prior to their participation. The study protocol was reviewed and approved by the IIT Delhi Institute Ethics Committee.

Additional information

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